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Gemma 4 Technical Report

Google DeepMind Releases Open-Weight Natively Multimodal Models from 2.3B to 31B Parameters, Including an Encoder-Free 12B Architecture Ingesting Raw Audio and Image Patches

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Google DeepMind’s Gemma 4 marks a substantial advance in open-weight multimodal language models, introducing a family spanning 2.3B to 31B parameters across dense and Mixture-of-Experts architectures. The flagship innovation is an encoder-free 12B model that ingests raw image patches and audio frames through the same token embedding space as text — eliminating the modality-specific encoder modules that have been a fixture of multimodal systems since their inception. All models include a built-in thinking mode and are released with full weights, quantized variants, and inference code.

AT A GLANCE

Problem: Open-weight multimodal models lag frontier closed models on STEM, long-context, and audio-visual tasks; separate encoder modules add architectural complexity and maintenance burden.

Why it matters: Open-weight multimodal models are essential for research reproducibility and on-device deployment; removing encoder modules simplifies the architecture fundamentally.

Method: Dense (2.3B, 9B), MoE (26B-A4B), and encoder-free (12B) architectures; thinking mode trained via SFT + RL; 4-bit QAT variants for on-device deployment.

Result: Gemma 4 12B (thinking) reaches 72.1% MMLU-Pro and 91.4% MATH-500; preferred over models 2–3× larger in human-rated evaluations.

Evidence: Evaluated on MMLU-Pro, MATH-500, MMBench, RULER (256K), and speech/audio captioning benchmarks.

The Big Picture

Open-weight language models have narrowed the gap with frontier closed models substantially over the past two years, but multimodal open-weight models have lagged: handling audio natively, processing very long contexts, and generating reasoning traces have each required separate architectural components with separate training pipelines.

Gemma 4 unifies these capabilities into a single model family with a consistent architecture and training procedure. The encoder-free 12B model is the most architecturally novel member: by treating image patches and audio frames as tokens in the same embedding space as text, it eliminates the need for a separately pretrained and maintained vision encoder and audio encoder, while retaining competitive performance on both modalities.

Why Existing Approaches Fall Short

Encoder-based multimodal models couple a visual or audio encoder (pretrained independently on large corpora) to a language model through a projection layer. This approach requires maintaining two (or three) separate model components with different training schedules, different fine-tuning protocols, and different quantization sensitivities. Performance is also bounded by the encoder’s representational quality: if the encoder is undertrained on certain input types (e.g., audio-visual correspondence), no amount of language model fine-tuning can recover the missing information.

Thinking modes in prior open-weight models have typically been implemented via supervised fine-tuning alone, without RL stabilization, leading to trace inflation — models generating arbitrarily long reasoning traces that do not improve answer quality. Gemma 4 addresses this with a constrained RL stage that optimizes final-answer correctness under a trace-length penalty.

Core Mechanism

The encoder-free architecture tokenizes image patches and audio frames using a learned patchification layer that maps each patch directly to the model’s embedding dimension:

$$\mathbf{e}_i = W_p \cdot \text{vec}(p_i) + \mathbf{b}_p$$

where p_i is the i -th patch (image or audio), W_p is the patchification weight matrix, and $\mathbf{b}_p$ is a bias. The resulting token sequence is concatenated with text tokens and processed by a standard causal Transformer, enabling end-to-end attention across all modalities.

Technical Formulation

The MoE model (26B-A4B) uses top- k expert routing with $k = 2$ across $E = 64$ experts. For a given hidden state $\mathbf{h}$, routing scores are computed as:

$$\mathbf{g}(\mathbf{h}) = \text{softmax}(\mathbf{h}W_g) \in \mathbb{R}^E$$

and the top-2 experts are selected and combined:

$$\text{MoE}(\mathbf{h}) = \sum_{i \in \text{top-2}(\mathbf{g})} g_i(\mathbf{h}) \cdot E_i(\mathbf{h})$$

Load balancing is enforced via an auxiliary loss:

$$\mathcal{L}_{\text{aux}} = \alpha \sum_{i=1}^E f_i \cdot P_i$$

where f_i is the fraction of tokens routed to expert i and P_i is the mean routing probability to expert i . The thinking mode RL stage applies correctness reward only to final-answer tokens:

$$\mathcal{L}_{\text{RL}} = -\mathbb{E}[R_{\text{correct}} \cdot \log \pi_{\theta}(a_{\text{answer}} \mid c, \tau)] + \beta \cdot \mathbb{E}[|\tau|]$$

where τ is the reasoning trace, $|\tau|$ is trace length, and β is the trace-length penalty.

Learning Procedure

1. **Pretraining:** All models are pretrained on diverse multimodal corpora including web text, code, image-caption pairs, and audio-speech pairs.
2. **SFT:** Models are fine-tuned on instruction-following datasets and on synthetically generated reasoning traces for thinking mode initialization.
3. **RL:** RLHF with correctness reward for factual and mathematical tasks, combined with the trace-length penalty to prevent trace inflation.
4. **QAT:** Quantization-aware training at 4-bit to produce on-device variants with minimal quality loss.

Experimental Findings

Benchmark	Gemma 4 12B	Previous best
MMLU-Pro	72.1%	68.4%
MATH-500 (thinking)	91.4%	88.7%
MMBench	81.6	79.1
RULER 128K	97.2%	94.8%
RULER 256K	87.1%	—
Speech recog. (WER)	4.3%	4.5%

The 4-bit QAT variants retain 97–99% of full-precision benchmark scores. Human-rated evaluations prefer Gemma 4 9B and 12B over models with 2–3× more parameters in head-to-head blind comparisons on open-ended generation tasks.

Ablations and Interpretation

Encoder-free vs. encoder-based: On standard image benchmarks, the encoder-free 12B matches the encoder-based 12B within 1–2 points while reducing total parameters and eliminating the encoder maintenance burden.

Thinking mode cost-benefit: Enabling thinking mode doubles average response latency but improves accuracy by 5–15 points on multi-step reasoning tasks. It is implemented as an opt-in feature controlled by the user prompt.

Trace length penalty: Without the trace-length penalty in RL, models generate traces 3–5× longer with no improvement in final-answer accuracy — confirming that unconstrained thinking mode RL leads to trace inflation rather than deeper reasoning.

MoE vs. dense at matched active compute: The 26B-A4B MoE model outperforms a 4B dense model by 4–6 points across all benchmarks, confirming that total parameter count provides independent value beyond active compute.

Reference

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